

Sheet1

Assessment	Main Course Concepts Addressed	Conceptual Depth	Technical Depth	Cognitive Load	Dependency on Earlier Work	Comments on Role in Course Progression
Homework 1	Entity extraction, 2NF design, ER diagrams, conceptual schema design, constraint discussion	Moderate	Low	Moderate	Builds foundation; design references reused in HW2 & HW3	Establishes fundamental database thinking; first exposure to formal modeling
Homework 2	SQL DDL, DML, inserts, deletes, prerequisite support tables, assistive SQL queries	Moderate–High	Moderate–High	Moderate	Directly uses HW1 schema	First technical implementation task; teaches schema creation and logic extraction
Homework 3	Multi-table JOINS, SQL reporting queries, prerequisite validation logic, detection of rule violations	High	High	Moderate–High	Requires HW1 & HW2 schema and data	Reinforces relational query fluency; prepares students for SQL components of the project
Homework 4	NoSQL concepts (key-value, document stores), limitations of relational systems, caching strategies, schema flexibility, performance considerations	High	Low–Moderate (research-based)	Moderate	Conceptually linked to HW1–3; no technical reuse required	Introduces CAP-aligned thinking and non-relational architectures; complements project and prepares for Week 13 topics
Midterm Exam (2025)	Analysis of historical case study, complex normalization, ERD extension, SQL creation, multi-step schema evolution, SQL queries, update logic, reasoning about relationships	Very High	Very High	Very High	Requires full command of HW1/HW2 skills	Serves as comprehensive assessment; integrates modeling, SQL, iteration, and logical reasoning across scenario